

Call & Response MIDI (v2.0.0)

Viral Samples | Intelligent Algorithmic Phrase Generator & 303-Style Sequencer

Formats: VST3, Audio Unit (AU), Standalone | macOS Universal (Mojave Intel & Apple Silicon) & Windows 64-bit

Attributions & Acknowledgements

Call & Response MIDI stands on the shoulders of creative musical innovators:

- **Nordl3**: Deep gratitude and attribution to **Nordl3** for the original project concept and pioneering the algorithmic foundations of automated dual-phrase musical dialogue.
- **SideBrain**: Sincere appreciation and attribution to **SideBrain** for his renowned Max for Live implementation, creative workflow concepts, and dedication to sharing innovative composition tools with the music production community.

Evolution by Viral Samples (v2.0.0)

Viral Samples evolved and expanded these foundational ideas into a native, high-performance C++/JUCE instrument for macOS and Windows:

- Full **Audio Unit (AU)**, **VST3**, and **Standalone** 64-bit binaries.
- **Universal macOS Binary**: Native compatibility with macOS 10.14 Mojave Intel (`x86_64`) through macOS Sonoma/Sequoia & Apple Silicon (`arm64` : M1, M2, M3, M4).
- **Native Windows 64-bit Binary**: Fully self-contained portable VST3 and Standalone executable with static runtime linking.
- **Brand New Ableton-Inspired Interface**: Beautiful dark theme with high-contrast elements, fixed aspect-ratio scaling (`1180 \times 750`), and ergonomic workflow.
- **Interactive Top Ribbon Bar**: Direct value display boxes for `OCT` , `KEY` , `SCALE` , `b/#` , `STEPS` , `RATE` , and `SWING` with dual dropdown & stepper controls.
- **Tempo-Relative Playback RATE**: Choose between `Half speed` , `0` (Normal tempo sync), and `Double speed` .
- **Continuous SWING & VELOCITY Dragging**: Intuitive vertical click-and-drag for seamless groove swing (`0%-100%`) and velocity (`1-127`).

- **Dual Phrase Lanes:**
- Independent **MUTE** bypass buttons for Call and Response.
- Dedicated per-lane **DICE** mutators with modifiable **MUTATE xx%** depth.
- Per-lane **LOCK** padlocks protecting phrases from global regenerations.
- Color-coded visual harmony (Orange for Call, Teal for Response).
- **Four-Tab Inspector Panel:**
- **GENERATE:** Style (40 styles), Motif (24 types), Strategy (38 types), Mode, and Ending with individual vector lock padlocks.
- **SHAPE:** Gate Length ($0.25\times-2.0\times$), continuous click-and-drag Velocity ($1-127$), Resolve, and Drive.
- **ARTICULATION:** Accent Boost, Slide Time, Filter Cutoff ($60\,\text{Hz}-14\,\text{kHz}$), Resonance, and Decay Time.
- **OPTIONS:** Routing (Channels 1–16, Combined / Split / Layered) and Timing (Sync Mode: Timeline, Restart, Free).
- **Enhanced Piano Roll:** Repeatable right-edge gate resizing up to $4.0\times$ note length, step auditioning with automatic mouse release stop, pitch transposition, and note editing.
- **Integrated 303 Analog Synth:** Resonant low-pass filter, fast envelope decay, and saturation drive for immediate zero-latency previews.
- **8 Rapid Preset Slots (A–H):** Instant **Cmd+Click** saving, **Double-Click** loading, and right-click context menus.
- **Multi-Level Undo/Redo & Panic:** 24-step non-destructive history and global All-Notes-Off MIDI Panic button.
- **Drag & Drop MIDI Export:** Export generated phrases directly into DAW tracks.

Feature Summary

Category	Highlights
Musical Engine	35 Scales (Diatonic, Pentatonic, Blues, Exotic, Arp Pools), 40 Musical Styles, 38 Response Strategies, 24 Opening Motif types.
Transposition	Real-time automatic note conforming on Key/Scale/Octave change. Preserves modal intervals and harmonic integrity.

Category	Highlights
Top Ribbon	Direct value displays for <code>OCT</code> , <code>KEY</code> , <code>SCALE</code> , <code>b/#</code> , <code>STEPS</code> , <code>RATE</code> , <code>SWING</code> with dropdown menus and stepper arrows.
Performance	Continuous SWING (<code>0\%-100\%</code>) and VELOCITY (<code>1-127</code>) click-and-drag, tempo-relative RATE (Half, 0, Double), Density, Octave Jump.
Phrase Lanes	Independent Mute, Dice, modifiable Mutate %, Lock padlocks, Octave transpose, Inversion, Reversal, Copy, and Channel routing.
Piano Roll	Color-coded note blocks matching lane themes, repeatable right-edge gate resizing up to <code>4.0\times</code> , step auditioning auto-stop.
Inspector	4 tabs (GENERATE, SHAPE, ARTICULATION, OPTIONS) with individual parameter locks for targeted procedural composition.
303 Synth	Built-in analog-modelled sawtooth/square oscillator, resonant 24dB low-pass filter, and snappy decay envelope.
Presets & History	8 quick-recall slots (A–H) with save/load indicators, 24-step undo/redo queue, global MIDI Panic button.
MIDI Export	Drag & Drop directly into DAW arrangements with Combined, Call Only, Response Only, or Layered multi-channel routing.
Compatibility	macOS Universal (Mojave 10.14+ Intel & Apple Silicon M1–M4); Windows 10/11 (64-bit VST3 & Standalone).

Installation & Setup

macOS Universal Installation (Recommended)

Double-click `Call & Response MIDI v2.0.0 Installer.pkg` to install:

- VST3 Plugin: `/Library/Audio/Plug-Ins/VST3/Call & Response MIDI.vst3`
- Audio Unit (AU): `/Library/Audio/Plug-Ins/Components/Call & Response MIDI.component`

- **User Documentation:** `/Library/Audio/Documentation/Viral Samples/Call & Response MIDI/`

The installer automatically de-quarantines the files, applies code signatures, sets POSIX permissions, and flushes `AudioComponentRegistrar` so DAWs (Logic Pro, Ableton Live, FL Studio, Studio One, Cubase, Reaper) recognize the plugin immediately.

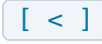
Windows Installation

Run `Call & Response MIDI v2.0.0 Windows Setup.exe`:

- **VST3 Plugin (64-bit):** `C:\Program Files\Common Files\VST3\Call & Response MIDI.vst3`
- **Standalone App:** `C:\Program Files\Viral Samples\Call & Response MIDI\Call & Response MIDI.exe`
- **Documentation:** `C:\Program Files\Viral Samples\Call & Response MIDI\Documentation\`

Quick Reference / Key Commands

Action	Gesture / Shortcut
Save Preset to Slot A–H	<code>Cmd+Click</code> (or <code>Shift+Click</code> / <code>Alt+Click</code> / <code>Ctrl+Click</code>) on slot button
Load Preset from Slot A–H	<code>Double-Click</code> on slot button
Preset Context Menu	<code>Right-Click</code> on slot button
Continuous Swing Drag	Click and drag vertically on <code>SWING</code> display box
Continuous Velocity Drag	Click and drag vertically on <code>VELOCITY</code> display in Inspector
Add / Delete Note in Piano Roll	<code>Double-Click</code> on grid cell or existing note
Move Note (Pitch & Step)	Click note center and drag horizontally or vertically

Action	Gesture / Shortcut
Resize Note Gate Duration	Hover right edge of note ( cursor) and drag horizontally
Audition Step	Click note (audition stops automatically when mouse is released)
Audition Pitch	Click piano keys along the left edge of the Piano Roll
Toggle Accent / Slide	Right-Click note for Accent (); Alt-Click note for Slide ()
Mute Lane	Click  on CALL or RESPONSE lane
Lock Lane	Click padlock icon on CALL or RESPONSE lane header
Parameter Lock / Unlock	Click padlock icon next to Style, Strategy, Motif, Mode, or Ending in Inspector
Offset Notes	Click rectangular  or  buttons under Left Panel rotaries